LAB ASSIGNMENT -3

1. Write a program to implement (a) pointer to an object (b) this pointer. Practice both '.' (dot operator) and '→' (arrow operator).

   Code:

   a)
```
#include<iostream>
using namespace std;

class Student
{
   int roll;

   public:

   void input()
   {
      cin >> roll;
   }

   void display()
   {
      cout << "Roll = " << roll << endl;
   }
};

int main()
{
   Student s;

   s.input();

   s.display();

   Student *p;

   p = &s;

   p->display();
}
```

   b)
```
#include<iostream>
using namespace std;

class Test
{
   int x;

   public:
```

```
    void setValue(int x)
    {
        this->x = x;
    }

    void display()
    {
        cout << "Value = " << this->x;
    }
};

int main()
{
    Test t;

    t.setValue(100);

    t.display();
}
```

2. Write a program to swap private values of two classes using a friend function.

Code:

```
#include<iostream>
using namespace std;

class B;

class A
{
    int x;

    public:

    A()
    {
        x = 10;
    }

    friend void swap(A &, B &);

    void display()
    {
        cout << "x = " << x << endl;
    }
};

class B
```

```
{
    int y;

    public:

    B()
    {
        y = 20;
    }

    friend void swap(A &, B &);

    void display()
    {
        cout << "y = " << y << endl;
    }
};

void swap(A &a, B &b)
{
    int temp;

    temp = a.x;
    a.x = b.y;
    b.y = temp;
}

int main()
{
    A a;
    B b;

    cout << "Before Swap" << endl;

    a.display();
    b.display();

    swap(a, b);

    cout << "After Swap" << endl;

    a.display();
    b.display();
}
```

3. Write a program to add data objects of two different classes using friend functions.

   Code:

```
#include<iostream>
using namespace std;
```

```
class B;

class A
{
    int x;

    public:

    A()
    {
        x = 10;
    }

    friend void add(A, B);
};

class B
{
    int y;

    public:

    B()
    {
        y = 20;
    }

    friend void add(A, B);
};

void add(A a, B b)
{
    cout << "Sum = " << a.x + b.y;
}

int main()
{
    A a;
    B b;

    add(a, b);
}
```

4. Write a program to demonstrate the working of friend class.

   Code:

```
#include<iostream>
using namespace std;
```

```
class A
{
    int x;

    public:

    A()
    {
        x = 100;
    }

    friend class B;
};

class B
{
    public:

    void display(A a)
    {
        cout << "Value = " << a.x;
    }
};

int main()
{
    A a;

    B b;

    b.display(a);
}
```

5. Write a program using Array of Objects to display area of multiple rectangles.

Code:

```
#include<iostream>
using namespace std;

class Rectangle
{
    int length, breadth;

    public:

    void input()
    {
        cin >> length >> breadth;
    }
```

```
    int area()
    {
        return length * breadth;
    }
};

int main()
{
    Rectangle r[3];

    for(int i=0; i<3; i++)
    {
        r[i].input();
    }

    for(int i=0; i<3; i++)
    {
        cout << "Area = " << r[i].area() << endl;
    }
}
```

6. Write a program to define function ***cube()*** as inline for calculating cube of a number.

   Code:

```
#include<iostream>
using namespace std;

inline int cube(int x)
{
    return x * x * x;
}

int main()
{
    int n;

    cin >> n;

    cout << "Cube = " << cube(n);
}
```

7. Write a program to pass an object as an argument and return the object from a function.
   a. Use pass-by-value
   b. Use pass-by reference

   Code:

```
a) #include<iostream>
using namespace std;
```

```
class Test
{
    int x;

    public:

    void input()
    {
        cin >> x;
    }

    void display()
    {
        cout << x << endl;
    }

    friend Test add(Test, Test);
};

Test add(Test t1, Test t2)
{
    Test temp;

    temp.x = t1.x + t2.x;

    return temp;
}

int main()
{
    Test t1, t2, t3;

    t1.input();
    t2.input();

    t3 = add(t1, t2);

    t3.display();
}
```

b)
```
#include<iostream>
using namespace std;

class Test
{
    int x;

    public:

    void input()
```

```
    {
        cin >> x;
    }

    void display()
    {
        cout << x << endl;
    }

    friend Test add(Test&, Test&);
};

Test add(Test &t1, Test &t2)
{
    Test temp;

    temp.x = t1.x + t2.x;

    return temp;
}

int main()
{
    Test t1, t2, t3;

    t1.input();
    t2.input();

    t3 = add(t1, t2);

    t3.display();
}
```